ROADWATCH · OXHACK

User Guide

How to install, run and use every RoadWatch feature

Contents

1. What RoadWatch is, in one page 2
2. Installation and startup 2
   - 2.1 Open the project directory 2
   - 2.2 Install dependencies (only if needed) 2
   - 2.3 Launch the dashboard 3
   - 2.4 Check that it started correctly 3
   - 2.5 Stop the dashboard safely 3
   - 2.6 Common installation and startup problems 3
3. Dashboard layout 4
   - 3.1 Sidebar controls 5
4. Inspection worklist & map: a complete worked example 5
   - Step 1: choose the week and capacity 5
   - Step 2: read the map and the ranked worklist 6
   - Step 3: search and filter 8
   - Step 4: inspect one block's supporting evidence 9
   - Step 5: download the worklist 9
   - Step 6: what an inspection recommendation means in practice 10
   - Step 7: try a different capacity or list 11
5. The other views explained 11
   - 5.1 Why a higher number is not automatically better 11
   - 5.2 Decision-policy comparison 12
   - 5.3 Weekly volume & scenarios 15
   - 5.4 Evidence & limitations 16
   - 5.5 Data provenance 18
6. Practical workflows 18
7. Feature and limitation reference 20
   - 7.1 Interactive dashboard features 20
   - 7.2 Command-line and data-processing capabilities (not in the dashboard) 21
   - 7.3 Experimental, unavailable or future features 21
8. Troubleshooting and frequently asked questions 22
   - 8.1 Problems and fixes 22
   - 8.2 Frequently asked questions 23
9. Glossary 24

1. What RoadWatch is, in one page

RoadWatch ranks Brooklyn's 23,448 street blocks by how likely each is to receive a **311 pothole report in the next 7 days**. It then turns the ranking into a weekly **inspection worklist** of a size you choose. The dashboard lets you explore that worklist on a map, see why each block is listed, compare alternative inspection policies, and review how well the method performed on 67 historical test weeks.

Keep four facts in mind whenever you read a number:

1. **It replays history.** Every week in the dashboard (7 July 2025 to 12 October 2026) is a saved forecast evaluated against what really happened. Nothing is live.
2. **It predicts reports, not defects.** A high rank means a public pothole report is relatively likely. It is not proof of a pothole, and nothing shown is a confirmed failure, a prevented failure or a saving.
3. **Scores are ranks, not probabilities.** Use them to order blocks within one week only.
4. **Some numbers depend on definitions.** Many reports name an intersection, so "did the list catch the report?" has several honest answers (§5.2).

What works offline: everything except the grey street basemap behind the map, which loads from the internet (CARTO). Downloading new data (`ingest`) also needs internet.

2. Installation and startup

2.1 Open the project directory

Open **Windows PowerShell** and go to the project:

```
cd C:\Users\drish\Downloads\Roadwatch
```

All commands in this guide are run from this folder. The project's Python lives in `.venv\` inside it. You do not need to "activate" anything: call it by its path, `.venv\Scripts\python.exe`.

Check it works:

```
.venv\Scripts\python.exe --version          # expect: Python 3.12.15
```

2.2 Install dependencies (only if needed)

The environment is already installed. You only need this if `.venv` is missing or broken, or you moved the project to a new computer. It uses `uv` (check with `uv --version`):

```
uv venv .venv --python 3.12
uv pip install --python .venv\Scripts\python.exe -r requirements-lock.txt
uv pip install --python .venv\Scripts\python.exe --no-deps -e .
```

`requirements-lock.txt` pins every package version, so the results stay reproducible.

2.3 Launch the dashboard

```
.venv\Scripts\python.exe -m streamlit run app.py --server.headless true --server.port 8501
```

After a few seconds the window prints:

```
  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://…:8501
```

Open **http://localhost:8501** in your browser. With `--server.headless true` the browser is not opened automatically. The flag also skips Streamlit's first-run e-mail question; without it, the first launch may wait at an `Email:` prompt, which you can dismiss by pressing **Enter**.

On shared Wi-Fi, add `--server.address localhost` so only your own computer can open the dashboard. The console then shows a single `URL: http://localhost:8501`.

2.4 Check that it started correctly

- **The browser shows** the title "*RoadWatch: Brooklyn pothole-report inspection prioritisation*", a yellow banner, and the metrics **Blocks to inspect 235** and **Capacity scenario 1%**.
- **Or from a second PowerShell window:** `(Invoke-WebRequest -UseBasicParsing http://localhost:8501/_stcore/health).Content` returns `ok`.

2.5 Stop the dashboard safely

Click the PowerShell window that runs it and press **Ctrl+C** (or close that window). This is always safe: the dashboard only *reads* saved files and never changes data, models or results.

2.6 Common installation and startup problems

Message or symptom	Cause	Fix
`The term '.venv\Scripts\python.exe' is not recognized`	You are not in the project folder	`cd C:\Users\drish\Downloads\Roadwatch`
`No module named streamlit` / `No module named roadwatch`	Environment incomplete	Run the three install commands in §2.2
`Port 8501 is already in use`	A previous dashboard is still running	Use it, stop it (Ctrl+C in its window), or start with `--server.port 8502` and open `http://localhost:8502`
Launch waits at `Email:`	First run without `--server.headless true`	Press **Enter**, or relaunch with the flag
Page shows "A required saved output is missing: …"	A precomputed output file is absent	Run the command printed in the message (§8.1)
`uv` not recognised	uv not installed or not on PATH	Install uv from its official site, or reuse the existing `.venv`

Message or symptom	Cause	Fix
Browser shows "This site can't be reached"	Dashboard not started, or wrong port	Start it (§2.3) and check the port in the URL
Yellow `use_container_width` warnings in the console	Streamlit deprecation notice	Harmless with the pinned Streamlit 1.66.1; ignore

3. Dashboard layout

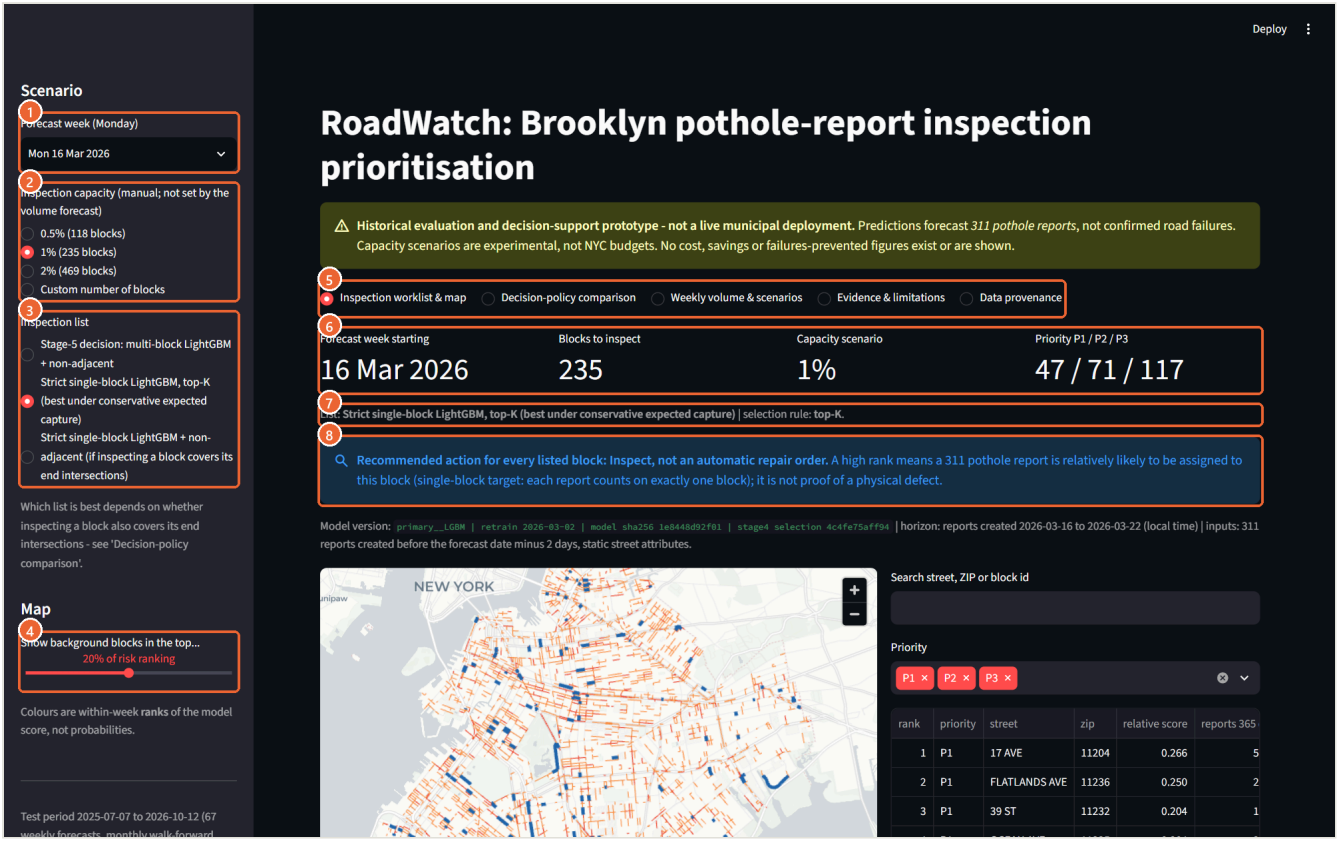


Figure 1. Main controls (week 16 Mar 2026). (1) forecast week, (2) inspection capacity, (3) inspection list, (4) map background slider, (5) view selector, (6) headline metrics, (7) selected list and selection rule, (8) recommended action.

The screen has two parts:

- **The sidebar on the left** holds the scenario controls. They apply to every view.
- **The main area** holds one of five **views**, chosen with the round buttons under the banner (5):

View	What it is for
Inspection worklist & map	The weekly list: map, ranked table, reasons, downloads
Decision-policy comparison	How alternative lists perform under four capture definitions
Weekly volume & scenarios	Weekly report counts and the (unreliable) volume forecast
Evidence & limitations	Test-period metrics, uncertainty, and what the numbers mean
Data provenance	Which data snapshots everything is built from

Links that open a specific state. You can add `?week=YYYY-MM-DD` and `?view=map|policy|volume|evidence|data` to the address. For example, `http://localhost:8501/?week=2026-03-16&view=policy` opens the policy view for 16 March 2026. An invalid or unavailable week shows a blue message in the sidebar and falls back to the latest week (Figure 15).

3.1 Sidebar controls

Control	Where	What it does	How to use it	Notes
Forecast week (Monday) (1)	Sidebar, top	Chooses which saved weekly forecast to show	Click the box, scroll or type, pick a Monday	67 weeks: Mon 07 Jul 2025 – Mon 12 Oct 2026. Default: the latest
Inspection capacity (2)	Sidebar	Number of blocks on the list: **0.5% (118 blocks)**, **1% (235 blocks)**, **2% (469 blocks)** or **Custom number of blocks**	Click an option. *Custom* shows **Inspection slots this week** (1–2,000, steps of 5; type a number and press Enter)	Capacity is a manual choice. The volume forecast never sets it. Custom lists are computed on demand (a few seconds) with the same engine
Inspection list (3)	Sidebar	Which ranking and selection rule builds the list	Click one of three options	Default: **Strict single-block LightGBM, top-K**. See §5.2 before switching
Show background blocks in the top… (4)	Sidebar, under *Map*	How much of the risk ranking is drawn behind the selected blocks: 1%, 5%, 20%, 50% or 100%	Drag the slider	Display only; it does not change the list. 100% draws all 23,448 blocks (slower)

The three inspection lists:

Option in the sidebar	Ranking	Selection rule	When it is the right choice
Strict single-block LightGBM, top-K (best under conservative expected capture), the default	strict single-block model	take the top K blocks	No assumption about what an inspection covers. Best under expected capture
Strict single-block LightGBM + non-adjacent (if inspecting a block covers its end intersections)	strict model	top K, skipping blocks that share an intersection with one already selected	Only if inspecting a block also covers potholes at its end intersections
Stage-5 decision: multi-block LightGBM + non-adjacent	multi-block model	non-adjacent	Kept for comparison. Its advantage depends on the same assumption, and the idea was proposed after test results were seen

4. Inspection worklist & map: a complete worked example

This walkthrough uses the saved forecast for **Monday 16 March 2026**, at the start of the March 2026 reporting surge. The values were checked against the running dashboard.

Step 1: choose the week and capacity

1. Open `http://localhost:8501/?week=2026-03-16&view=map`, or pick **Mon 16 Mar 2026** in *Forecast week (Monday)*.
2. Leave *Inspection capacity* on **1% (235 blocks)** and *Inspection list* on the default.

The metrics row (6) shows **16 Mar 2026 · 235 · 1% · 47 / 71 / 117**.

Metric	Meaning
Forecast week starting	The Monday the forecast covers (reports created that Monday to Sunday)
Blocks to inspect	List length (K)
Capacity scenario	The option chosen in the sidebar
Priority P1 / P2 / P3	List split: P1 = top 20% of the list, P2 = next 30%, P3 = the rest. For 235 blocks: 47 / 71 / 117

Under the metrics:

- **List and selection rule (7):** "selection rule: **top-K**".
- **Recommended action (8):** *Inspect, not an automatic repair order.*
- **Model version caption:** for example `primary_LGBM | retrain 2026-03-02 | model sha256 1e8448d92f01 | stage4 selection 4c4fe75aff94`. It names the monthly model that produced this week's ranking and the frozen selection record.

Step 2: read the map and the ranked worklist

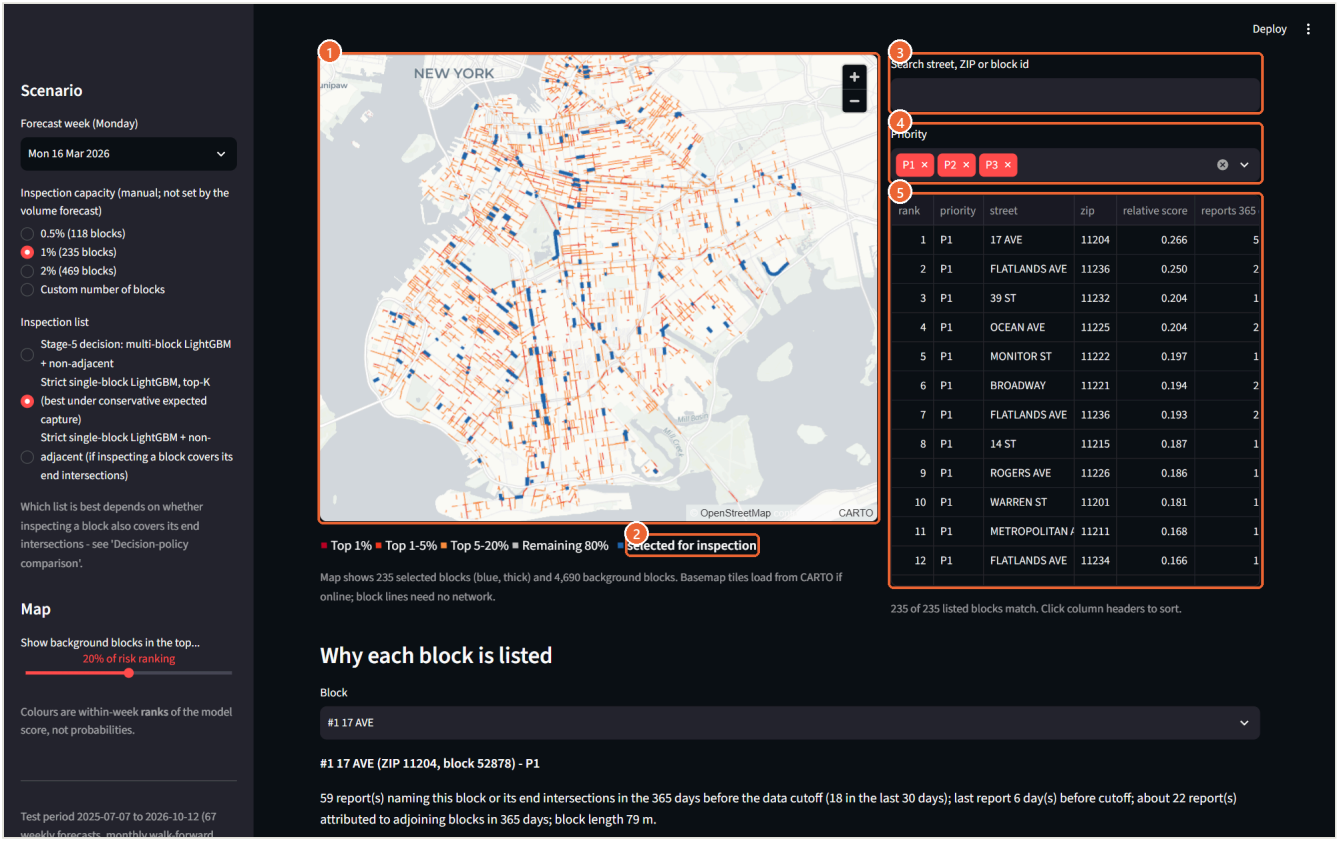


Figure 2. (1) map, (2) legend, (3) search box, (4) priority filter, (5) ranked worklist.

The map (1):

- **Lines:** thick **blue** lines are the 235 blocks on the list. Thin coloured lines are the rest of the ranking (dark red = top 1% of all blocks, red = top 1–5%, orange = top 5–20%, grey = remaining 80%, shown only if the slider allows).
- **Navigation:** drag to pan, scroll or use **+ / −** to zoom.
- **Hover a selected block** to see "*Inspection #rank (priority), street, ZIP, reports in last 365 days*".
- **Hover a background block** to see its within-week risk rank and band.

- **Colours are ranks** within this week, not probabilities.

The worklist table (5):

Column	Meaning
rank	1 = highest-ranked block this week
priority	P1 / P2 / P3 tier within the list
street, zip, block_id	Location (block id = CSCL `physicalid`)
relative score	The model's raw ranking score. Comparable **only within the same week and list**; **not a probability**
reports 365 d, reports 30 d	Reports naming this block (or one of its candidate locations) in the year or month before the data cut-off
days since last	Days between the most recent such report and the cut-off (capped at 365)
length m, travel_lanes	Road attributes from the street centreline

For 16 Mar 2026 the top rows are:

Rank	Priority	Street	ZIP	Relative score
1	P1	17 AVE	11204	0.266
2	P1	FLATLANDS AVE	11236	0.250
3	P1	39 ST	11232	0.204

Sorting and the table toolbar:

- Click a column header to sort, and again to reverse.
- Hovering over the table shows a small toolbar: **Show/hide columns**, **Download as CSV** (exactly what is displayed), **Search** and **Fullscreen**.
- The official worklist exports are the two download buttons in Step 5. They contain more columns.

Step 3: search and filter

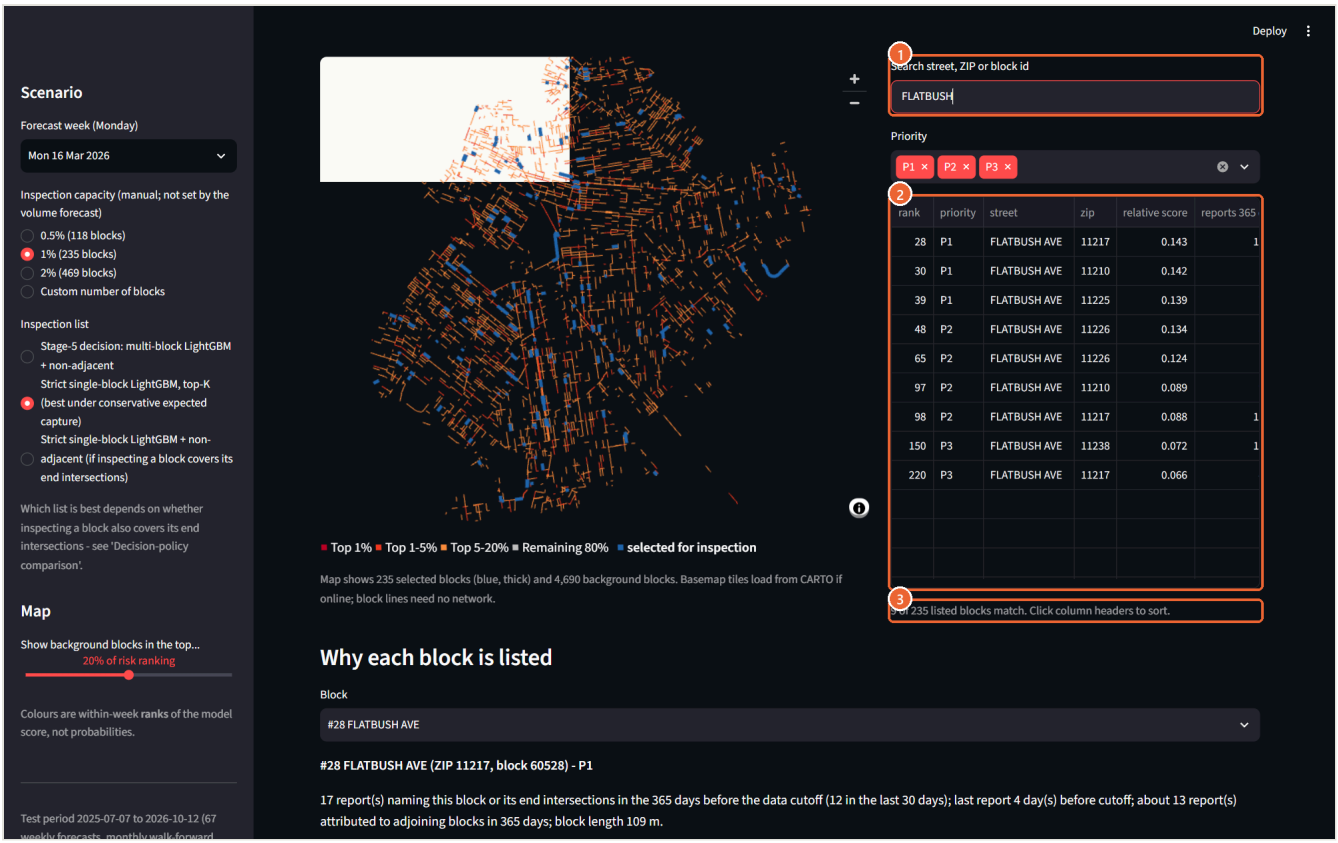


Figure 3. Typing FLATBUSH (1) filters the list (2): "9 of 235 listed blocks match" (3).

- **Search street, ZIP or block id** matches any part of the street name, ZIP or block id (not case-sensitive). Example: `FLATBUSH` gives 9 blocks; `11204` gives the blocks in that ZIP.
- **Priority** keeps only the tiers you leave selected. Remove a tier with its ×.
- **Filters only change the table.** The map, the metrics and the downloads always show the full list.
- If nothing matches, the caption says "0 of 235 listed blocks match" and the *Block* selector below shows "-".

Step 4: inspect one block's supporting evidence

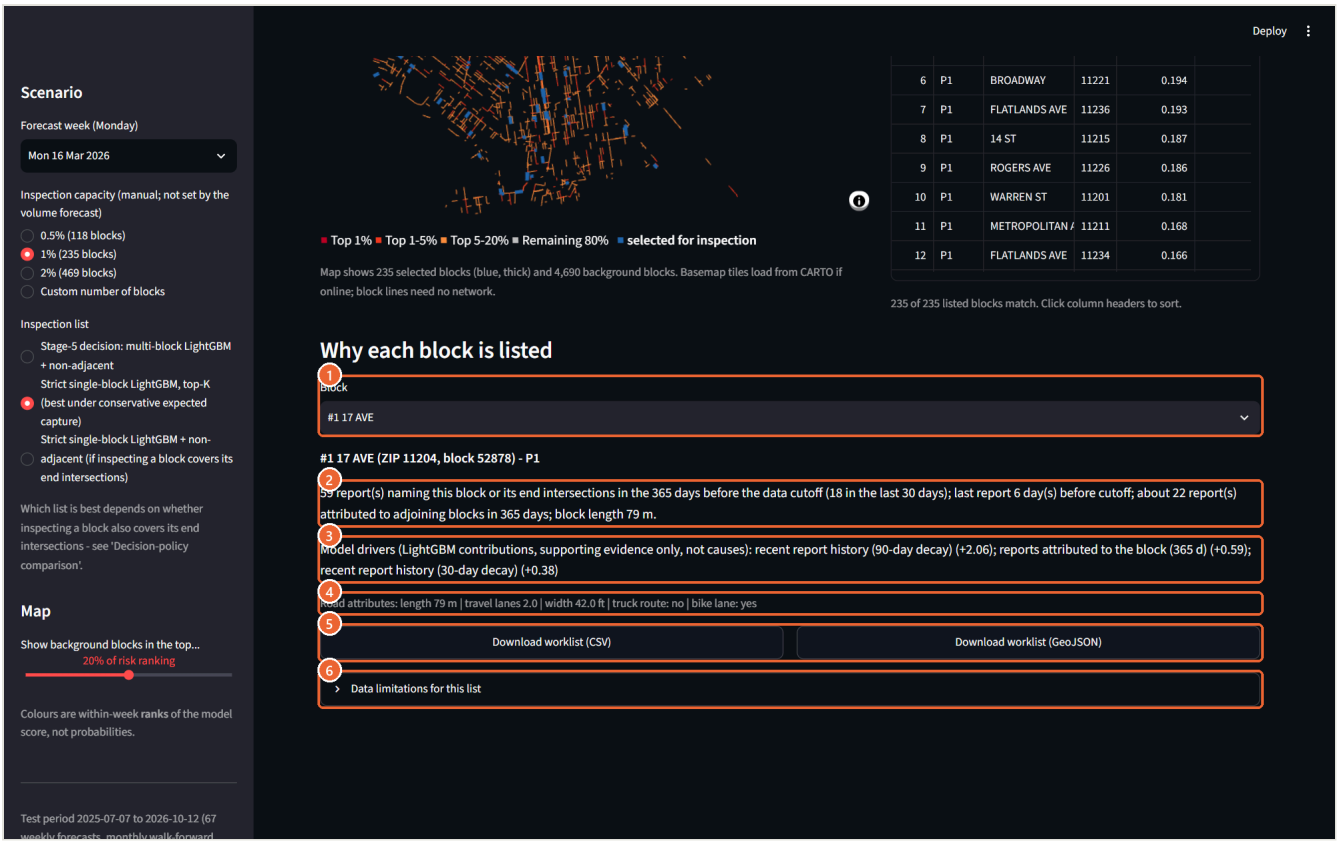


Figure 4. Why each block is listed: (1) block selector, (2) report history, (3) model drivers, (4) road attributes, (5) downloads, (6) data limitations.

Scroll to **Why each block is listed** and pick a block in **Block** (1). The selector offers the blocks currently shown in the table. For #1 17 AVE you see:

- **Header:** *#1 17 AVE (ZIP 11204, block 52878) – P1.*
- **Report history (2):** "59 report(s) naming this block or its end intersections in the 365 days before the data cutoff (18 in the last 30 days); last report 6 day(s) before cutoff; about 22 report(s) attributed to adjoining blocks in 365 days; block length 79 m."
- **Model drivers (3):** the three features that pushed this block's score up most, for example "recent report history (90-day decay) (+2.06)". They explain the *model*. They are **supporting evidence, not causes** of potholes.
- **Road attributes (4):** length 79 m, travel lanes 2.0, width 42.0 ft, truck route no, bike lane yes.

What you can conclude: the block is listed mainly because of a dense recent report history, here and nearby. **What you cannot conclude:** that a pothole exists now, how severe it is, or that repairing it would prevent anything.

Step 5: download the worklist

The two buttons (5) save the **whole list** for the selected week, capacity and list:

Button	File name	Contents
Download worklist (CSV)	`roadwatch_worklist_2026-03-16_strict_topk_1pct.csv`	235 rows × 24 columns

Button	File name	Contents
Download worklist (GeoJSON)	`roadwatch_worklist_2026-03-16_strict_topk_1pct.geojson`	235 line features with the same 24 properties

The file name records the week, the list (`strict_topk`, `strict_nonadj` or `multi_nonadj`) and the capacity (`0.5pct`, `1pct`, `2pct`, or `K100` for a custom 100). Files go to your browser's Downloads folder.

The 24 columns:

- `forecast_week_start`, `priority`, `rank`, `block_id`, `full_street_name`, `zip`, `model_score_relative`;
- `reports_365d`, `reports_30d`, `days_since_last_report`, `adjacent_block_reports_365d`;
- `length_m`, `travel_lanes`, `street_width_ft`, `truck_route_type`, `bike_lane`;
- `rationale`, `model_drivers`, `recommended_action`, `capacity_scenario`, `selection_rule`, `model_version`;
- `mid_lat`, `mid_lon` (block midpoint, WGS84).

The GeoJSON adds each block's line geometry.

Using the files:

- **CSV:** opens in Excel, Google Sheets or pandas. Sort by `rank`; use `mid_lat`/`mid_lon` for routing or mapping tools.
- **GeoJSON:** opens directly in QGIS (*Layer → Add Layer → Add Vector Layer*) or ArcGIS.
- **Web tools:** sites such as geojson.io also open it, but **uploading to a website sends the file to that service**. The data are derived from public sources, but check before sharing externally.

Step 6: what an inspection recommendation means in practice

The list says: *these blocks are where next-week pothole reports are relatively most likely; send an inspector to look.* An inspector would visit each block, record what they find, and only then decide on repairs. RoadWatch has **no data on what inspectors find**, so it cannot say how many listed blocks actually have potholes (§6, *Reviewing model performance*).

Step 7: try a different capacity or list

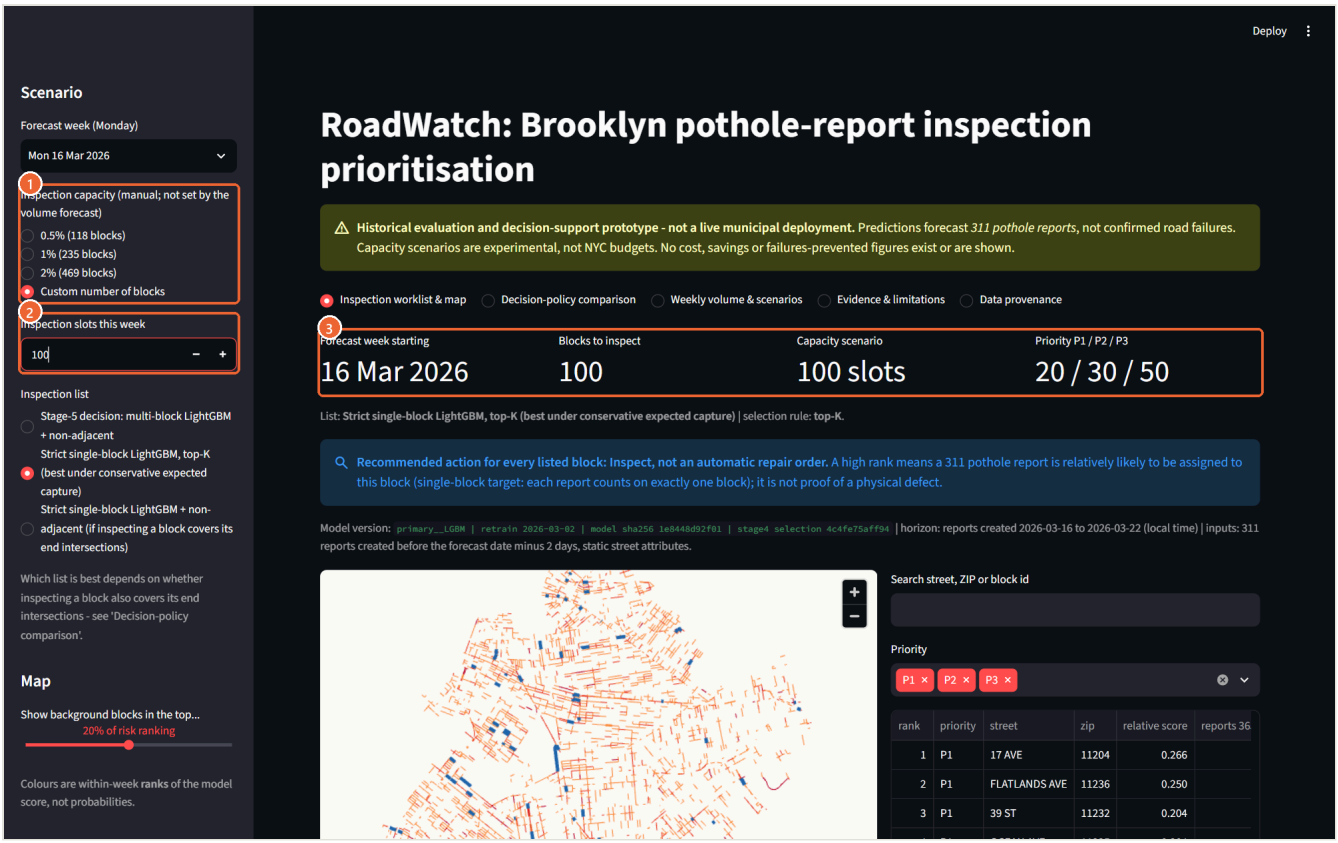


Figure 5. Custom capacity: choose "Custom number of blocks" (1), type 100 in "Inspection slots this week" (2) and press Enter; the metrics update (3) to 100 blocks, 20 / 30 / 50.

Change	Effect for 16 Mar 2026 (verified)
2% (469 blocks)	Blocks to inspect **469**; P1/P2/P3 **94 / 141 / 234**
0.5% (118 blocks)	**118**; **24 / 35 / 59**
Custom 100	**100**; **20 / 30 / 50**; download names end in `K100`
Inspection list → **Stage-5 decision: multi-block LightGBM + non-adjacent**	Caption changes to "selection rule: **top-K + non-adjacent (validation-decided)** – a block is skipped if it shares an intersection node with a block already selected"; the blue lines spread out

The **Data limitations for this list** expander (Figure 4, 6) restates the target and the caveats for whichever list you selected.

5. The other views explained

5.1 Why a higher number is not automatically better

Two lists can only be compared if they are measured against **the same definition** of success.

Example. In *Decision-policy comparison*, the column **Precision vs own target** reads:

List	Precision vs own target	Target
Strict single-block LightGBM (top-K)	0.095	primary

List	Precision vs own target	Target
Historical-rate baseline	0.117	multi_block
Plain top-K multi-block LightGBM	0.164	multi_block

It looks as if the baseline (0.117) beats the strict model (0.095). It does not:

- The baseline is scored against the **multi-block** target, where a block counts if it is *any* of a report's possible locations. That is an easier test.
- Against the same strict target, the baseline's precision is **6.6%**, against the strict model's **9.5%** (*Evidence & limitations*, Stage 4 table).

Always check the target in brackets, or the capture definition, before comparing.

5.2 Decision-policy comparison

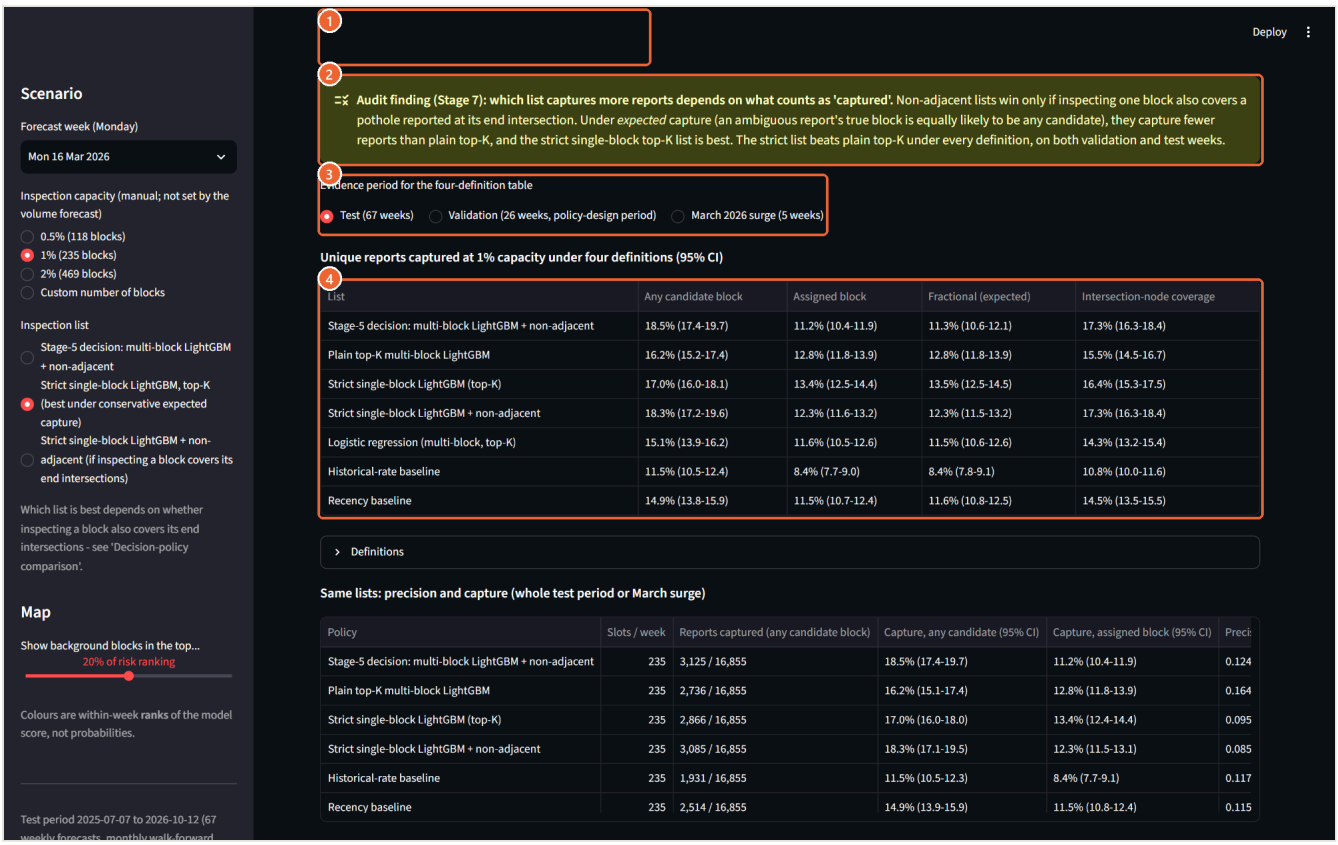


Figure 6. (1) Period for the lower table and charts, (2) audit finding, (3) evidence period for the four-definition table, (4) four-definition table.

What it shows. Each row is a saved inspection list, evaluated over many historical weeks at the capacity chosen in the sidebar. A custom capacity shows the 1% evidence, with a note.

The four-definition table (4) gives the share of next-week unique reports captured, with a 95% interval in brackets. Choose the weeks with **Evidence period for the four-definition table** (3):

Option	Weeks	Purpose
Test (67 weeks)	2025-07-07 → 2026-10-12	Main result
Validation (26 weeks, policy-design period)	2025-01-06 → 2025-06-30	Where the lists were designed

Option	Weeks	Purpose
March 2026 surge (5 weeks)	stress test	Few weeks: wide intervals

The four columns differ in how a report with several candidate blocks is credited:

Column	A report counts as captured if…	Read it as
Any candidate block	any of its possible blocks is inspected	optimistic upper view
Assigned block	its one assigned block is inspected	estimate of expected capture
Fractional (expected)	share of its candidate blocks inspected	expected value; **the default's best case**
Intersection-node coverage	intersection reports count if any adjoining block is inspected	valid **only if** an inspection covers the block's end intersections

Worked example (Test, 1%):

List	Fractional (expected)	Intersection-node coverage
Strict top-K (default)	**13.5%** (12.5–14.5)	16.4%
Strict + non-adjacent	12.3%	**17.3%**
Stage-5 decision (multi + non-adjacent)	11.3%	**17.3%**
Historical-rate baseline	8.4% (7.8–9.1)	10.8%

Conclusion you can draw: with no assumption about inspection coverage, the default list is best. The non-adjacent lists win only if an inspection covers end intersections, which nobody has yet confirmed in the field. **Conclusion you cannot draw:** that any list finds 13.5% (or 17.3%) of potholes. These are *reports*.

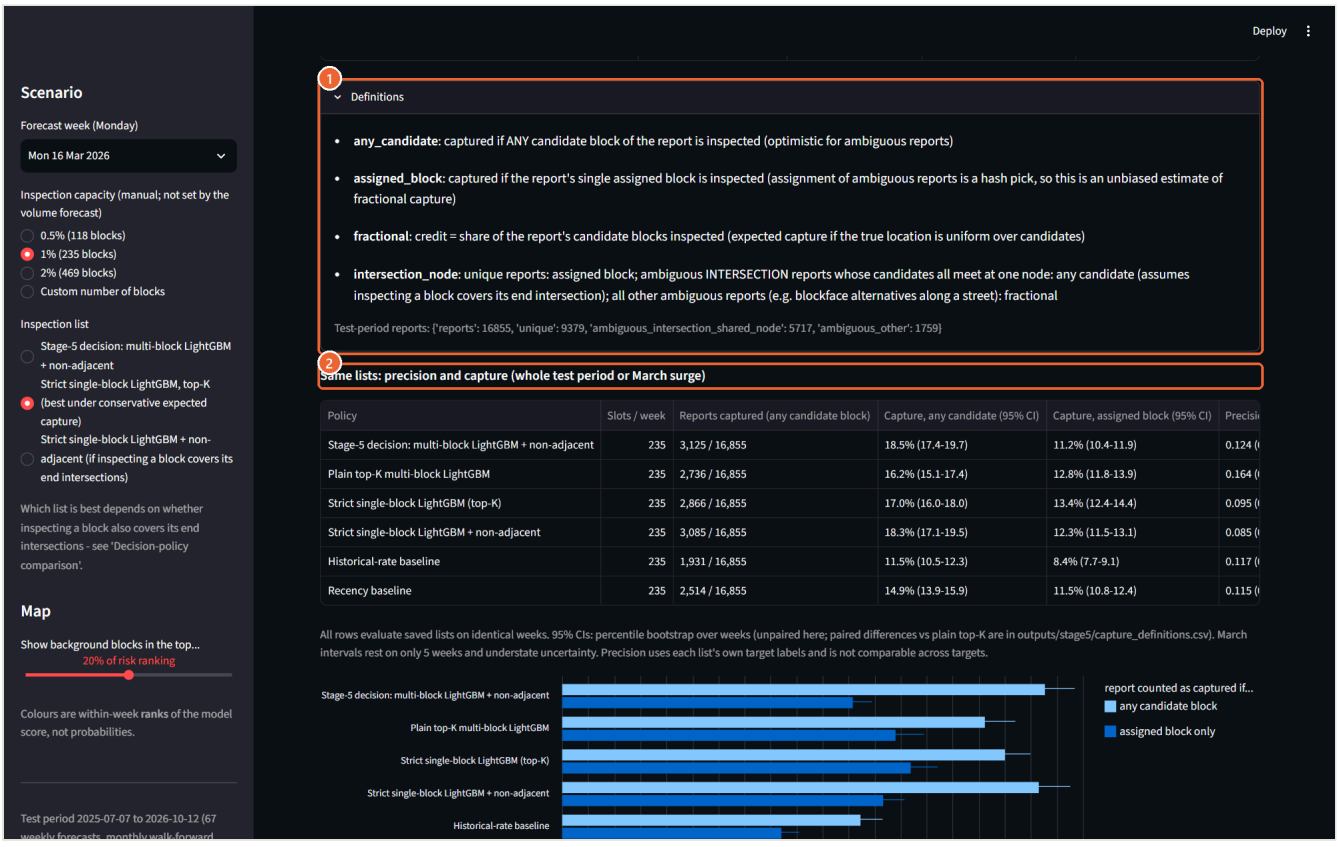


Figure 7. (1) Definitions expander (exact wording and the test-period report mix: 16,855 reports, 9,379 unique, 5,717 intersection, 1,759 other); (2) the precision and capture table for the period chosen in "Period".

The "Same lists" table (Figure 7, 2) is driven by the **Period** radio at the top: *Whole test period (67 weeks)* or *March 2026 surge (5 weeks)*.

Column	Meaning
Slots / week	K
Reports captured (any candidate block)	for example 2,866 / 16,855 for the default list
Capture, any candidate (95% CI) / assigned block (95% CI)	capture under two of the definitions
Precision vs own target (95% CI) [target]	share of listed blocks that got a report, against the list's own target (§5.1)
Reports per 100 inspections	reports captured (any candidate) per 100 listed blocks; 18.2 for the default list



Figure 8. (1) Capture by list under "any candidate block" (light) and "assigned block only" (dark), with 95% intervals; (2) "How to read this" explanation. Further down, a monthly line chart shows capture by month (hover for values).

Uncertainty intervals:

- **Method:** week-level bootstrap. Weeks are resampled 1,000 times and the middle 95% is reported.
- **Overlap:** overlapping intervals in this table do not prove two lists are equal. The paired differences in `outputs/stage5/capture_definitions.csv` are more precise.
- **March intervals** rest on only 5 weeks and understate uncertainty.

5.3 Weekly volume & scenarios



Figure 9. (1) The red box states plainly that the volume model did not beat simple baselines; (2) actual weekly reports (blue, from 2021) and the walk-forward forecast (orange dashed) with its 80% interval (shaded).

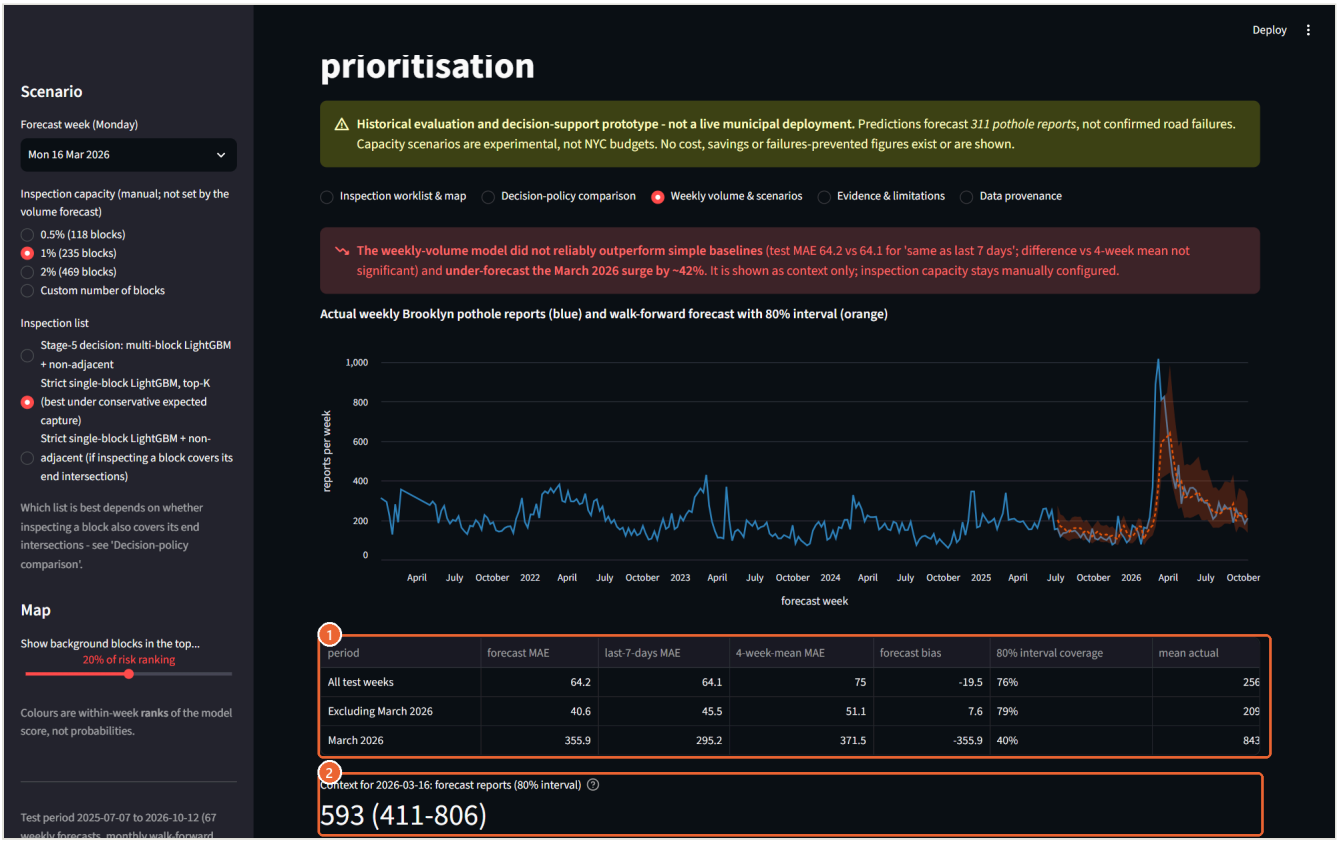


Figure 10. (1) Errors by period; (2) the forecast for the selected week.

How to read the table (reports per week; MAE = average absolute error):

Period	Forecast MAE	"Same as last 7 days" MAE	4-week-mean MAE	80% interval coverage	Mean actual
All test weeks	64.2	64.1	75.0	76%	256
Excluding March 2026	40.6	45.5	51.1	79%	209
March 2026	355.9	295.2	371.5	40%	843

What it tells you:

- In normal weeks the forecast is a little better than naive rules.
- In the March 2026 surge, reports ran about 4× normal (843 per week), the forecast fell about 42% short, and its 80% interval covered only 2 of 5 weeks.
- Over all weeks it is no better than "same as last week".
- The metric for 16 Mar 2026 shows **593 (411-806)** forecast reports.

Conclusion: use it as context only. This is why capacity is a manual choice. A pre-specified rule that raised capacity when the forecast was high lowered the reports found per inspection, from 19.8 to 17.0 per 100.

5.4 Evidence & limitations

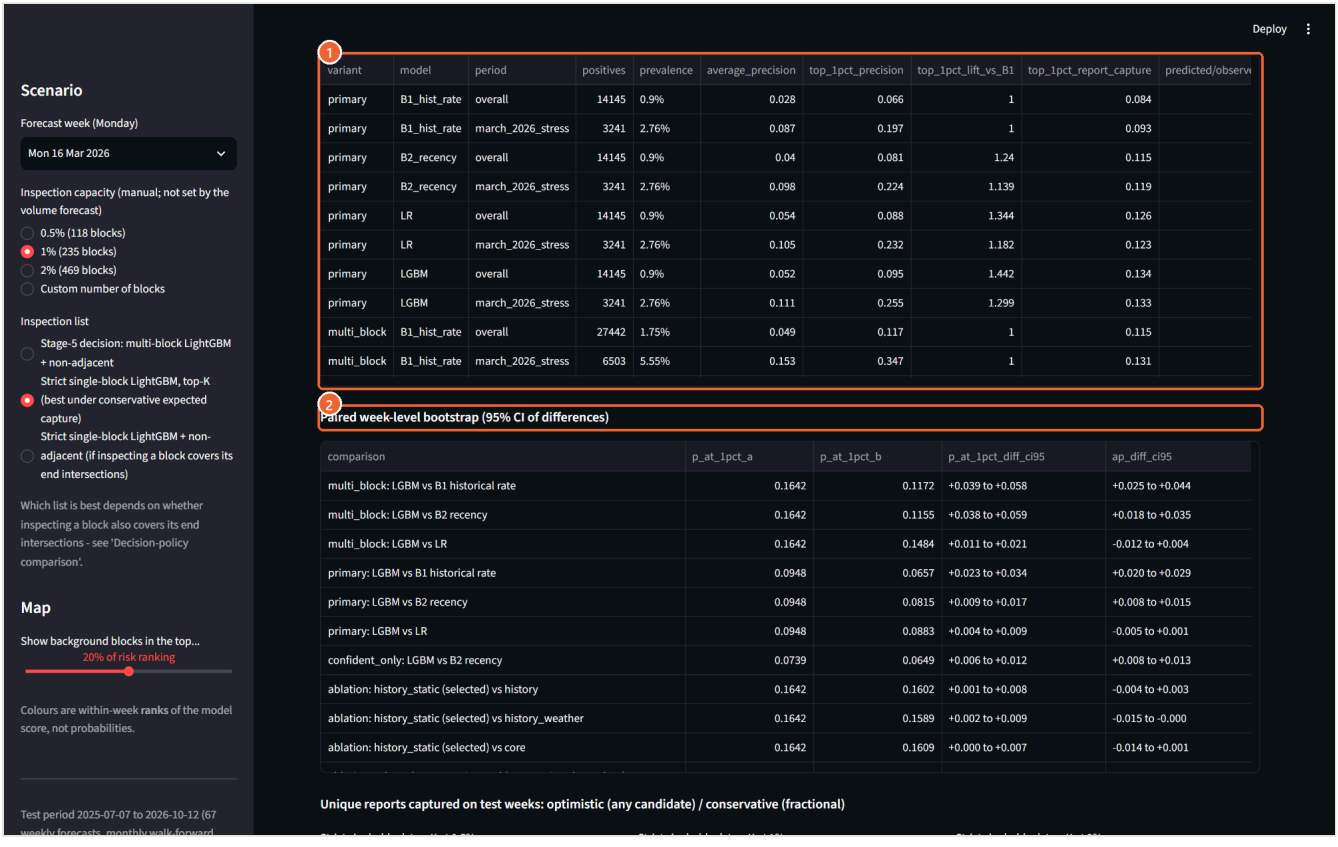


Figure 11. (1) Walk-forward test metrics per target, model and period; (2) paired week-level bootstrap of differences.

The Stage 4 table (1) has one row per target (`primary` = strict, `multi_block`), model and period (`overall` = 67 test weeks, `march_2026_stress` = the March 2026 weeks):

Column	Meaning
positives	positive block-weeks
prevalence	share of block-weeks with a report
average_precision	ranking quality over all blocks
top_1pct_precision	precision of a 235-block list
top_1pct_lift_vs_B1	precision ÷ historical-rate precision
top_1pct_report_capture	reports captured by the list on this target
predicted/observed volume	calibration in the large; 1 = right total

Worked example (strict target, overall):

Model	Precision	Report capture	Lift vs historical rate
LightGBM	9.5%	13.4%	1.44×
Historical rate	6.6%	8.4%	—

The bootstrap table (2) gives the 95% interval of the difference: for strict LightGBM against the historical rate, **+0.023 to +0.034** in precision (+2.3 to +3.4 points). The interval excludes zero, so the improvement is not a fluke of particular weeks.

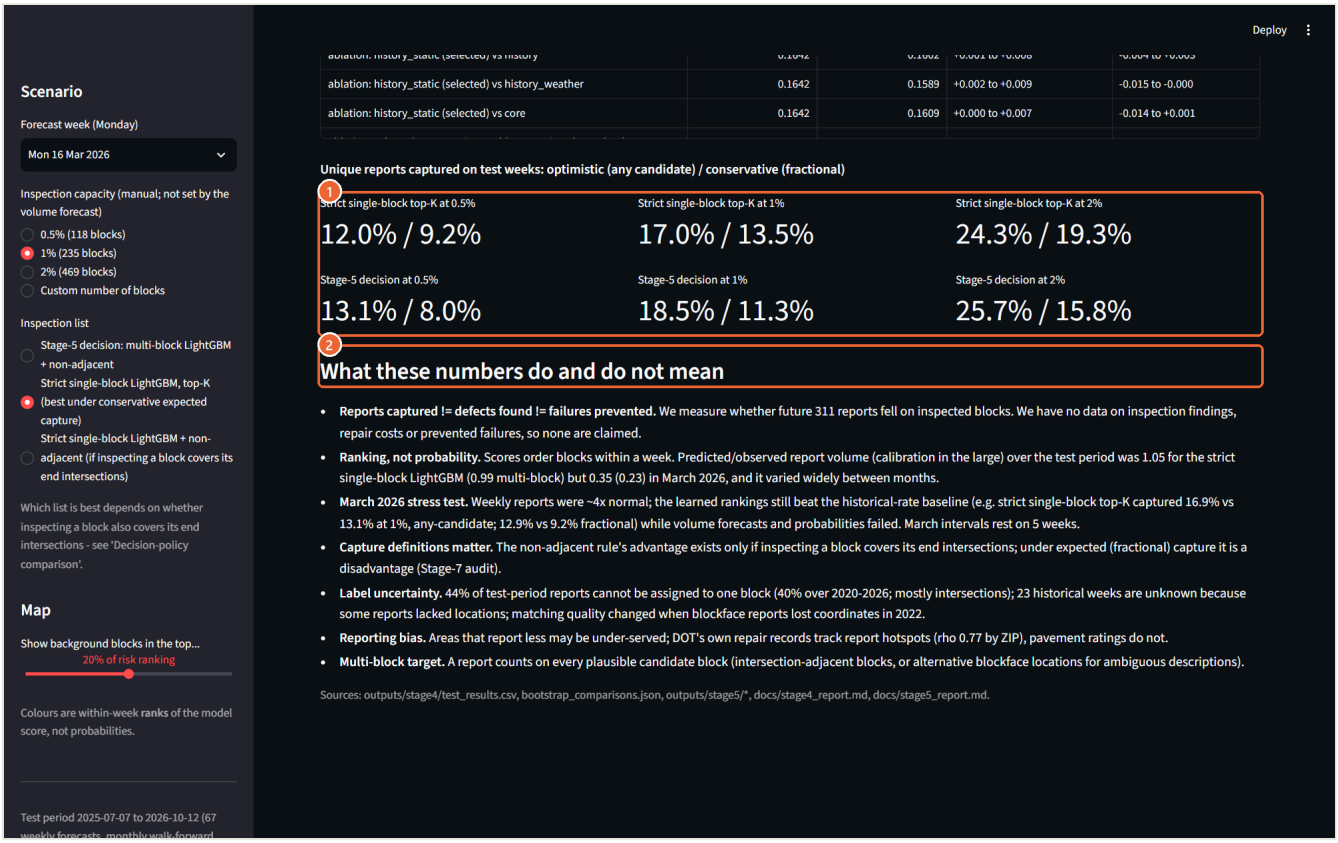


Figure 12. (1) Capture pairs "any candidate / fractional" for the default and Stage-5 lists at 0.5%, 1% and 2%; (2) plain-language limitations.

The metric pairs read as *optimistic / conservative*. For example, **Strict single-block top-K at 1%: 17.0% / 13.5%**. The bullets underneath summarise the limitations: reports are not defects, ranking not probability, the March surge, capture definitions, label uncertainty and reporting bias.

5.5 Data provenance

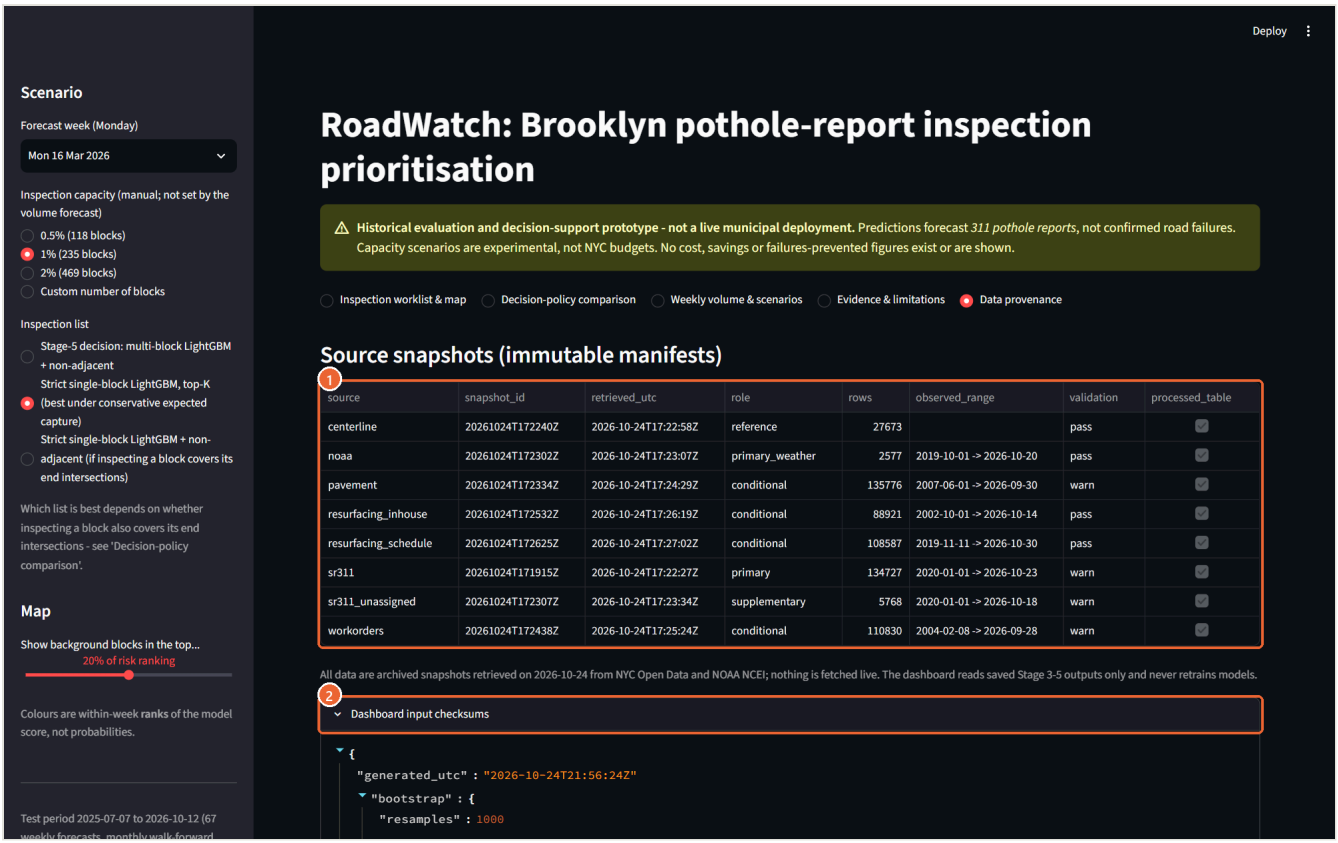


Figure 13. (1) Latest finalised snapshot per source; (2) checksums of the dashboard's own inputs.

Column	Meaning
source	sr311 (311 street-condition requests), sr311_unassigned, centerline, noaa, pavement, workorders, resurfacing_inhouse, resurfacing_schedule
snapshot_id, retrieved_utc	when the immutable copy was downloaded (all on 2026-10-24)
role	primary, supplementary, reference, primary_weather (tested, not selected), conditional (diagnostics only)
rows, observed_range	size and date span (for example sr311: 134,727 rows, 2020-01-01 → 2026-10-23)
validation	**pass**, or **warn** (non-fatal findings, documented in `outputs/data_quality/`). A **fail** would never be promoted
processed_table	✓ = a validated Parquet table exists

Source freshness: the data end on 23 October 2026 for 311 and 20 October 2026 for NOAA. Nothing is fetched live, so the dashboard does not get fresher on its own (§7).

6. Practical workflows

Internet marks which workflows need a connection: only the map background, and downloading new data.

#	Workflow	Steps	Internet?
1	**Explore a historical week**	1. Pick a week in *Forecast week (Monday)*. 2. Read the metrics and the map. 3. Scroll the table. 4. Open *Weekly volume & scenarios* to see whether that week was busy	No (basemap optional)
2	**Identify a block for closer inspection**	1. In the table, sort by **reports 30 d** or filter *Priority* to **P1**. 2. Pick the block in **Block** under *Why each block is listed*. 3. Read the history, drivers and attributes. 4. Hover it on the map to see neighbours	No
3	**Compare two inspection policies**	1. Open *Decision-policy comparison*. 2. Keep *Evidence period* on **Test**. 3. Compare **Fractional (expected)**, then **Intersection-node coverage**. 4. Repeat with **Validation**. 5. To see the lists themselves, switch *Inspection list* in the sidebar and return to the map	No
4	**Choose a fixed capacity scenario**	1. In *Evidence & limitations*, read the capture pairs at 0.5%, 1% and 2% (for the default: fractional 9.2% → 13.5% → 19.3%). 2. Weigh the extra reports against the extra inspections (118 → 235 → 469). 3. Set the capacity in the sidebar. Do **not** size it from the volume forecast	No
5	**Export a worklist for another application**	1. Set week, capacity and list. 2. Click **Download worklist (CSV)** or **(GeoJSON)**. 3. Open in Excel or QGIS (§4, Step 5). From the command line (writes to `outputs/stage5/worklists/`): `.venv\Scripts\python.exe -I -m roadwatch stage5 worklist --week 2026-03-16 --capacity 0.01 --policy primary__LGBM --rule topk`	No
6	**Review model performance**	1. *Evidence & limitations*: Stage 4 table, filtered mentally to `primary`, `overall`. 2. Bootstrap table: does the interval exclude 0? 3. *Decision-policy comparison*: capture under each definition. 4. *Weekly volume*: March 2026 row	No
7	**Check source freshness and provenance**	1. *Data provenance*: retrieved_utc, observed_range, validation. 2. From PowerShell: `.venv\Scripts\python.exe -I -m roadwatch status` (lists every snapshot and its state). 3. Detailed checks: `outputs/data_quality/data_quality_report.md`	No
8	**Diagnose a failed dashboard or missing output**	1. Read the PowerShell window running Streamlit (Python errors appear there). 2. If the page names a missing file, run the printed command. 3. Health check: `(Invoke-WebRequest -UseBasicParsing http://localhost:8501/_stcore/health).Content`. 4. Run `.venv\Scripts\python.exe -I -m pytest -q`: failing tests name the broken part. 5. See §8	No

Offline: only the grey street background disappears (Figure 14). Downloading new data (`ingest`) requires internet; see §7 before doing it.

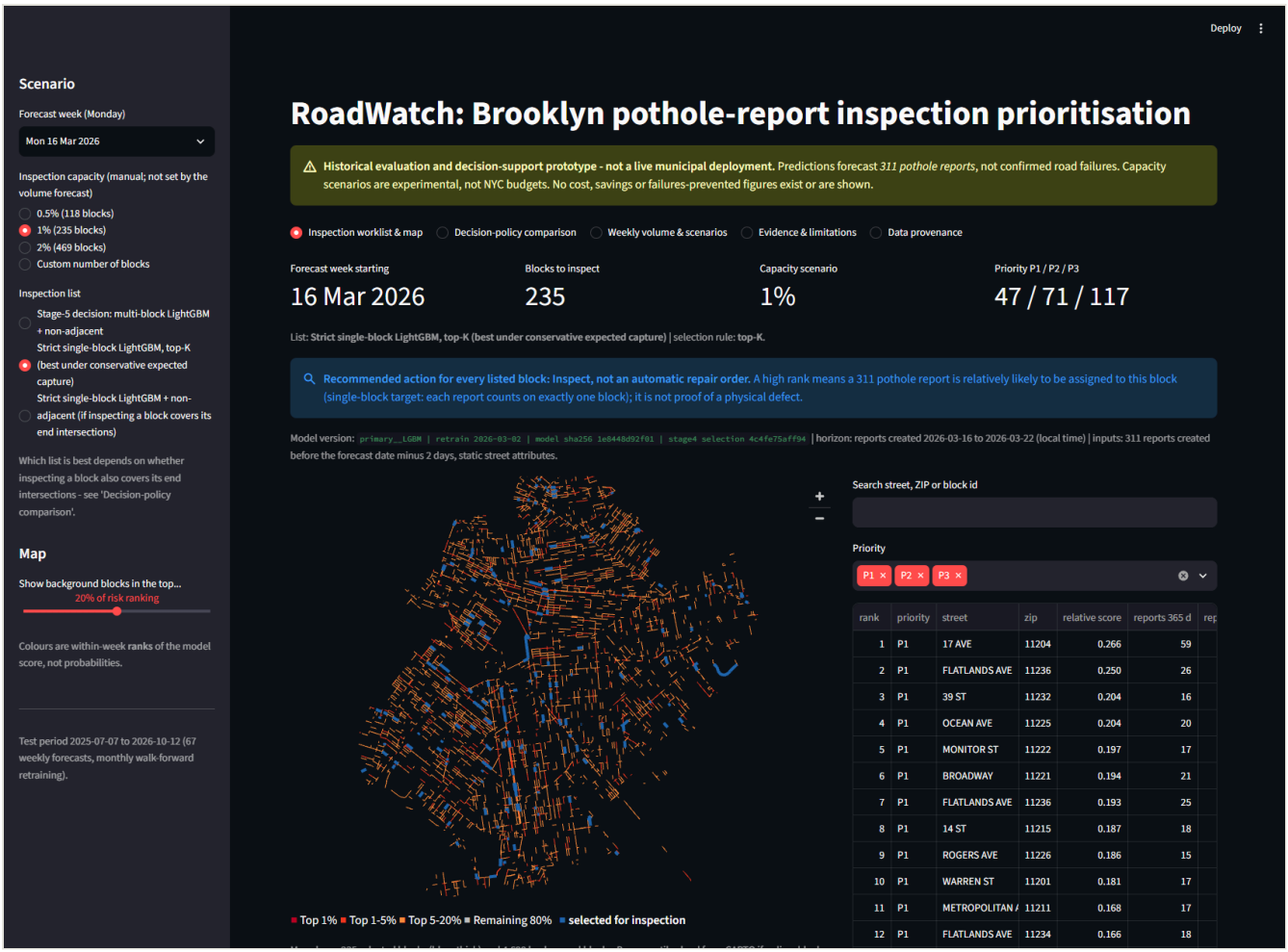


Figure 14. With all internet hosts blocked, the map still draws every block line; only the street basemap is missing.

7. Feature and limitation reference

7.1 Interactive dashboard features

Feature	Purpose	Inputs	Produces	Limitations
Forecast week selector	Choose a saved week	67 Mondays, 2025-07-07 → 2026-10-12	All views for that week	Historical only; no weeks after 12 Oct 2026
Capacity (0.5 / 1 / 2%)	List length	saved worklists	118 / 235 / 469 blocks	Experimental scenarios, not NYC budgets
Custom capacity	Any K from 1 to 2,000	saved predictions	list computed on demand	A few seconds per change; policy evidence exists only for 0.5/1/2%
Inspection list selector	Choose ranking and rule	three saved list types	map, table and downloads for that list	Non-adjacent lists depend on an unconfirmed coverage assumption
Map	Where the listed blocks are	block geometry, week ranks	interactive lines, hover tooltips	Basemap needs internet; colours are ranks
Background slider	Context density	week ranks	more or fewer background lines	Display only

Feature	Purpose	Inputs	Produces	Limitations
Search / priority filter	Find blocks	worklist	filtered table	Table only; map and downloads unaffected
Block evidence	Why a block is listed	worklist row	history, drivers, attributes	Drivers explain the model, not causes
CSV / GeoJSON download	Hand-off to other tools	worklist	files with 24 columns	Historical lists; no inspection outcomes
Policy comparison	Compare lists	saved evaluation files	four-definition and precision tables, charts	Definitions matter (§5.2); March has only 5 weeks
Weekly volume	Context on demand	saved volume forecasts	chart, error table, week forecast	Not reliable; not used for capacity
Evidence & limitations	Test results	saved Stage 4/5 files	tables, intervals, caveats	Read-only
Data provenance	Source status	snapshot manifests	snapshot table, checksums	No live refresh button

7.2 Command-line and data-processing capabilities (not in the dashboard)

Command (`.venv\Scripts\python.exe -I -m roadwatch …`)	What it does	Internet?	Caution
`status`	Lists all data snapshots	No	Safe
`report`	Writes the data-quality report	No	Safe
`stage5 worklist --week W --capacity C --policy P [--rule R] [--diversify]` (R = `topk` or `nonadjacent`)	One week's worklist to `outputs/stage5/worklists/` (CSV and GeoJSON)	No	Default policy is the Stage 5 multi-block list. For the dashboard's default use `--policy primary__LGBM --rule topk`
`stage6 build`	Recomputes the dashboard's precomputed inputs from saved outputs	No	Safe; deterministic
`ingest <sources>`	Downloads **new** data snapshots, validates and promotes them	**Yes**	Changes the provenance view, and later stages if re-run. **Do not run before a presentation**
`rebuild`, `stage3`, `stage4 …`, `stage5 …`	Rebuild the pipeline from archived snapshots	No	Long-running. Never re-run `stage4 select` (it would overwrite the frozen selection record)

7.3 Experimental, unavailable or future features

Item	Status
Non-adjacent lists	**Experimental comparison.** Depends on an unconfirmed inspection-coverage assumption
ZIP cap (`--diversify`)	**Experimental, command line only.** No measurable effect in tests

Item	Status
Weekly volume forecast	**Context only**; not reliable
Weather, pavement-rating and work-order features	**Tested or ingested, not used** by the selected model
Forecasts for weeks after 12 Oct 2026	**Unavailable.** Would require new data and new model runs; not packaged as a routine
Live 311 data, automatic refresh	**Unavailable** (future development)
Defects found, severity, costs, savings	**Unavailable.** Never measured; would need inspection outcomes and cost data

8. Troubleshooting and frequently asked questions

8.1 Problems and fixes

Problem	What you see	Fix
Unavailable forecast week	Blue sidebar message: "No saved prediction for … Showing 2026-10-12."	Choose a Monday between 7 Jul 2025 and 12 Oct 2026
Invalid date in the link	"'2031-02-30' is not a valid date; showing the latest available week 2026-10-12." (Figure 15)	Correct the `?week=` value (YYYY-MM-DD)
Missing map tiles	Block lines on a blank background	Expected without internet; everything else works
Map empty or not drawn	No lines at all	Reload the page; check the browser allows WebGL; the table and downloads still work
Empty worklist table	"0 of 235 listed blocks match"	Clear the search box; re-add P1/P2/P3 in *Priority*
Custom capacity seems stuck	Spinner "Selecting blocks with the Stage-5 engine…"	Wait a few seconds; press Enter after typing the number
Download does nothing	—	Check the browser's download bar and permissions; try another browser, or use **Download as CSV** in the table toolbar (displayed columns only)
"A required saved output is missing: `path`. Generate it with: `command`"	Red box	Run the printed command from the project folder, then reload
Failed data refresh (`ingest`)	Exit code 1 (core source failed) or 2 (optional source failed); error lines in the console	Read `outputs\logs\run_*.log`. The previous processed tables stay in use automatically. Re-run `ingest <source>` (it resumes), or `ingest <source> --new` to restart a snapshot. Rate limits: add `SOCRATA_APP_TOKEN` to `.env`
Numbers differ from this guide	—	Someone rebuilt outputs. Run `.venv\Scripts\python.exe -I -X utf8 scripts\final\verify_final.py`, which lists every changed number and its source file

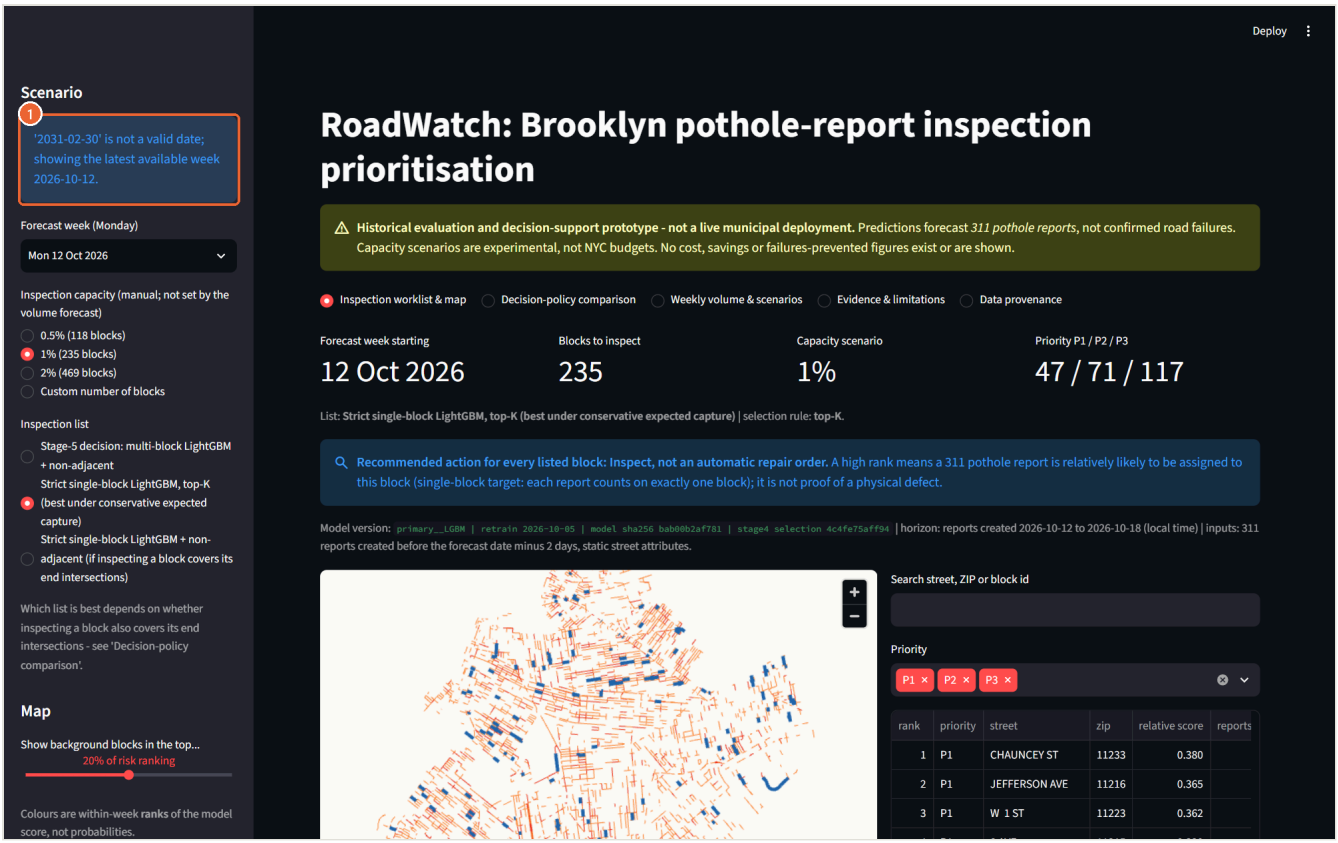


Figure 15. Invalid week in the link: the sidebar explains and shows the latest week instead.

8.2 Frequently asked questions

Why does the strict list "catch" fewer reports than the Stage-5 list in some tables? Under *any candidate* counting, a list that spreads blocks across intersections gets credit for every possible location of an intersection report. Under expected (fractional) counting the strict list is better (13.5% vs 11.3%). See §5.2.

What happened in March 2026? Weekly reports rose to about 843, roughly 4× normal. The ranking still beat the historical-rate baseline: 16.9% vs 13.1% (any candidate) and 12.9% vs 9.2% (expected), over only 5 weeks. But the volume forecast fell about 42% short, and the strict model's predicted/observed volume dropped to 0.35. Use ranks, not probabilities, in surges.

Why do the strict and multi-block numbers differ so much? They answer different questions:

- **Strict:** each report counts on one block. Test prevalence is 0.9%, and LightGBM precision 9.5%.
- **Multi-block:** each candidate block counts. Test prevalence is 1.7%, and precision 16.4%.

Only compare like with like (§5.1).

Is a relative score of 0.266 a 26.6% chance? No. It is a ranking score, comparable only within the same week and list.

Why can't I choose this week's real forecast? The dashboard replays saved historical forecasts (7 Jul 2025 – 12 Oct 2026). Producing new weeks needs new data and new model runs, which is future work (§7.3).

Does a block on the list have a pothole? Not necessarily. RoadWatch predicts *reports*. In 2025, citywide, 9.4% of pothole reports were closed as "not found" and 8.8% as duplicates.

9. Glossary

Term	Meaning
Block	A street segment between two intersections; the unit RoadWatch ranks (23,448 in Brooklyn)
Forecast week	Monday to Sunday after the forecast date; reports created in it are the target
Worklist	The ranked list of blocks to inspect in one week
Capacity (K)	Number of blocks on the list: 0.5%, 1%, 2% or custom
Priority P1/P2/P3	Top 20% / next 30% / rest of the list
Relative score	The model's raw ranking score; not a probability
Strict single-block target	Each report counts on exactly one block (the default)
Multi-block target	Each of a report's possible blocks counts
Candidate block	A block a report could refer to
Capture	Share of next-week reports that fell on listed blocks (four definitions, §5.2)
Precision	Share of listed blocks that received a report
95% interval (CI)	Range that would contain the value in 95% of resampled histories
Non-adjacent rule	Never list two blocks that meet at the same intersection
Walk-forward test	Replaying history: retrain monthly on the past, predict the next weeks
March 2026 surge	Five weeks with about 4× normal report volume; the main stress test
Snapshot	Immutable copy of a data download, with checksums
MAE	Mean absolute error of the weekly volume forecast